Executive Business Intelligence Dashboard

E-Commerce Sales & Customer Analytics

Project technologies:

- Python – Data Processing and Cleaning
- Pandas – Data Manipulation
- MySQL – Database Management and SQL Analytics
- Microsoft Power BI – Interactive Data Visualization

Project objective:

To clean, transform, analyze, and visualize e-commerce transaction data to identify sales trends, customer purchasing behaviour, product performance, and business insights.

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Problem Statement & Project Objectives

Problem Statement

E-commerce businesses generate large volumes of customer and transaction data. However, raw data may contain inconsistent formats, incomplete records, and information distributed across multiple files.

Without proper data processing and analysis, identifying meaningful business trends and making informed decisions becomes difficult.

Project Objectives

1. Clean and prepare customer transaction data for analysis.
2. Analyze customers based on age-independent available attributes such as customer ID, country, and purchase frequency.
3. Calculate total revenue, average order value, and purchase quantities.
4. Identify top-performing products and product categories.
5. Analyze monthly sales trends and order statuses.
6. Build an interactive Power BI dashboard to present business insights.

Expected Outcome

A structured analytical dataset and an executive dashboard that support data-driven business decisions.

Dataset Collection & Data Profiling

Dataset Source

Kaggle – E-Commerce Sales and Customer Analytics Dataset.

Source files and data overview

Dataset	Records	Purpose
customers.csv	300	Customer details
orders.csv	1,000	Order and status information
order_items.csv	2,000	Product-level transaction details
products.csv	50	Product and category information
sales_cleaned.csv	1,625	Completed transaction records

Dataset characteristics

- 300 unique customer records.
- 1,000 original orders.
- 2,000 item-level transaction records.
- 50 unique products.
- Five countries: Italy, Spain, Morocco, France, and Germany.
- Four product categories: Hair, Makeup, Body, and Skin.
- Transaction period: January–December 2024.

Order Status Distribution

Distribution of all 1,000 original orders from orders.csv.

Cancelled

Completed

Returned

Important: 2,000 represents item-level rows, not 2,000 distinct orders. The 1,625-row sales_cleaned.csv file represents completed transaction item records.

Data Cleaning & Preparation Using Python

Tools Used: Python, Pandas

Data Preparation Process

1. Data Loading

- Imported CSV files using Pandas.
- Loaded customer, order, product, and order-item datasets.

2. Data Inspection

- Checked dataset dimensions and column names.
- Examined data types and missing values.
- Identified duplicate records and inconsistent formats.

3. Data Cleaning

- Standardized column names and data formats.
- Checked missing and duplicate records.
- Converted date fields into appropriate date formats.
- Validated product IDs, customer IDs, and order IDs.

4. Data Transformation

- Merged customer, order, product, and order-item information.
- Calculated line-level revenue.

$$\text{Line Revenue} = \text{Quantity} \times \text{Price}$$

5. Feature Engineering

- Extracted year, month number, and month name.
- Created a Year-Month field for monthly analysis.

6. Data Export

- Generated the final processed file: sales_analytics.csv.

Final Processed Dataset

2,000 rows

16 columns

- Item-level analytical dataset.
- No missing values or duplicate rows reported in the final validation.
- Prepared for SQL analysis and Power BI visualization.

End-to-End Project Architecture

System Architecture

Use this exact workflow in your architecture diagram:

1. Data Sources

Customers • Orders • Order Items • Products

2. Python & Pandas

Data loading • Cleaning • Transformation • Validation

3. Processed Dataset

sales_analytics.csv — 2,000 item-level records

4. MySQL Database

executive_bi • Analytical tables • Six SQL views

5. CSV Analytical Exports

Six Power BI-ready analytical datasets

6. Microsoft Power BI

KPI cards • Charts • Filters • Interactive dashboard

7. Business Insights & Validation

Sales trends • Customer behaviour • Product performance

Architecture Explanation

Python handles data preparation, MySQL performs structured analytical queries, and Power BI converts the resulting datasets into interactive visual reports.

MySQL Database & SQL Analytics

Database Name: executive_bi

Main analytical tables:

- sales_analytics

- orders

Six SQL Analytical Views

SQL View	Purpose
monthly_sales_kpis	Monthly revenue, orders, units and AOV
order_status_kpis	Total, completed, cancelled and returned orders
category_performance	Category revenue and contribution
country_performance	Country-wise revenue and sales
product_performance	Product-level revenue and units
customer_performance	Repeat and one-time customer analysis

Key SQL Operations

- SELECT – Retrieve relevant records.
- JOIN – Combine related datasets.
- GROUP BY – Aggregate sales and customer information.
- SUM() – Calculate total revenue and quantities.
- COUNT(DISTINCT) – Count unique orders and customers.
- ROUND() – Format calculated KPI values.
- CASE WHEN – Classify customer types and order statuses.

Important Revenue Calculation

$$\text{Revenue} = \sum(\text{Quantity} \times \text{Price})$$

Result: Six analytical SQL views were created and exported as CSV files for Power BI.

Executive KPIs & Order Status Analysis

Key Performance Indicators (KPIs)

311,111

Completed Transaction Revenue

1,000

Total Original Orders

10.30%

Cancellation Rate

9.20%

Return Rate

Order Status Breakdown

Order Status	Number of Orders
Completed	805
Cancelled	103
Returned	92
Total	1,000

KPI Calculations

$$\text{Cancellation Rate} = \frac{103}{1000} \times 100 = 10.30\%$$

$$\text{Return Rate} = \frac{92}{1000} \times 100 = 9.20\%$$

Important Data Reconciliation Note

The source orders table contains 805 completed orders, but the processed item-level sales_analytics table contains only 695 distinct completed order IDs.

Therefore:

- Completed transaction revenue of 311,111 is calculated from the 695 represented completed orders.
- The 1,000 total orders and status percentages are calculated from the original orders table.
- The difference of 110 completed orders requires further investigation.

Do not combine these two different order counts as if they were the same measure.

Power BI Dashboard & Visualizations

Visualization Tools: Microsoft Power BI Desktop

Six Major Dashboard Visualizations

1. KPI Cards

- Completed transaction revenue.
- Total orders.
- Cancellation rate.
- Return rate.

2. Monthly Sales Revenue Trend

- Line chart.
- Tracks monthly completed revenue from January to December 2024.

3. Revenue by Product Category

- Bar or column chart.
- Compares Hair, Makeup, Body, and Skin.

4. Revenue by Country

- Bar or column chart.
- Compares performance across five countries.

5. Top 10 Products by Revenue

- Clustered bar chart.
- Displays the ten highest-revenue products.

6. Customer & Order Status Analysis

- Customer type chart comparing repeat and one-time customers.
- Order status donut chart, provided its three status categories are correctly labelled and validated.

Interactive Features

- Category slicer.
- Visual-level filtering.
- Dynamic chart updates.

- Cross-filtering between supported visuals.

Dashboard Objective

To provide a consolidated view of business performance through interactive visual analytics.

Key Business Findings & Insights

1. Monthly Sales Performance

- Highest monthly revenue: January — 30,948.
- Lowest monthly revenue: June — 22,819.
- Highest monthly completed order count in the analytical dataset: October — 70.
- Highest average order value: July — 574.26.

2. Product Category Performance

Category	Revenue	Contribution
Hair	101,637	32.67%
Makeup	81,085	26.06%
Body	76,011	24.43%
Skin	52,378	16.84%

Insight: Hair is the highest-revenue product category.

3. Country Performance

- Italy generated the highest completed revenue: 70,719.
- Spain recorded 69,897.
- Morocco recorded 65,164.

4. Product Performance

- Top-performing product: Product_16.
- Category: Makeup.
- Revenue: 9,732.
- Units sold: 135.

5. Customer Behaviour

Customer Type	Customers	Revenue
Repeat Customers	196	278,796
One-time Customers	71	32,315

Insight: Repeat customers represent approximately 73.4% of the 267 customers represented in completed item-level transactions and contribute approximately 89.6% of their completed transaction revenue.

Validation, Limitations & Conclusion

Data Validation

- Checked dataset dimensions and column types.
- Validated customer, order, and product identifiers.
- Checked missing and duplicate records.
- Verified revenue calculations.
- Compared SQL analytical outputs against the source data.
- Manually checked Power BI KPI and chart values.

Important Project Limitations

1. The source order dataset contains 1,000 orders, while only 861 distinct order IDs appear in the item-level dataset.
2. The item-level dataset represents 695 completed orders, compared with 805 completed orders in the original order table.
3. The reason for the 110-order difference has not yet been established.
4. Currency information is not specified in the available dataset.
5. Product cost information is unavailable; therefore, profit and profit margin cannot be calculated.
6. Further validation of the order-status donut chart is necessary if its labels and segments have not been verified.

Future Enhancements

- Investigate missing order-item relationships.
- Develop automated ETL pipelines.
- Implement predictive sales forecasting.
- Apply customer segmentation and recommendation models.
- Deploy the dashboard to a cloud-based BI environment.

Conclusion

The project successfully demonstrates an end-to-end business intelligence workflow using Python, Pandas, MySQL, and Power BI. It transforms raw e-commerce transaction data into structured analytical datasets, SQL-based KPIs, and interactive visualizations that reveal sales performance, product trends, and customer purchasing behaviour.